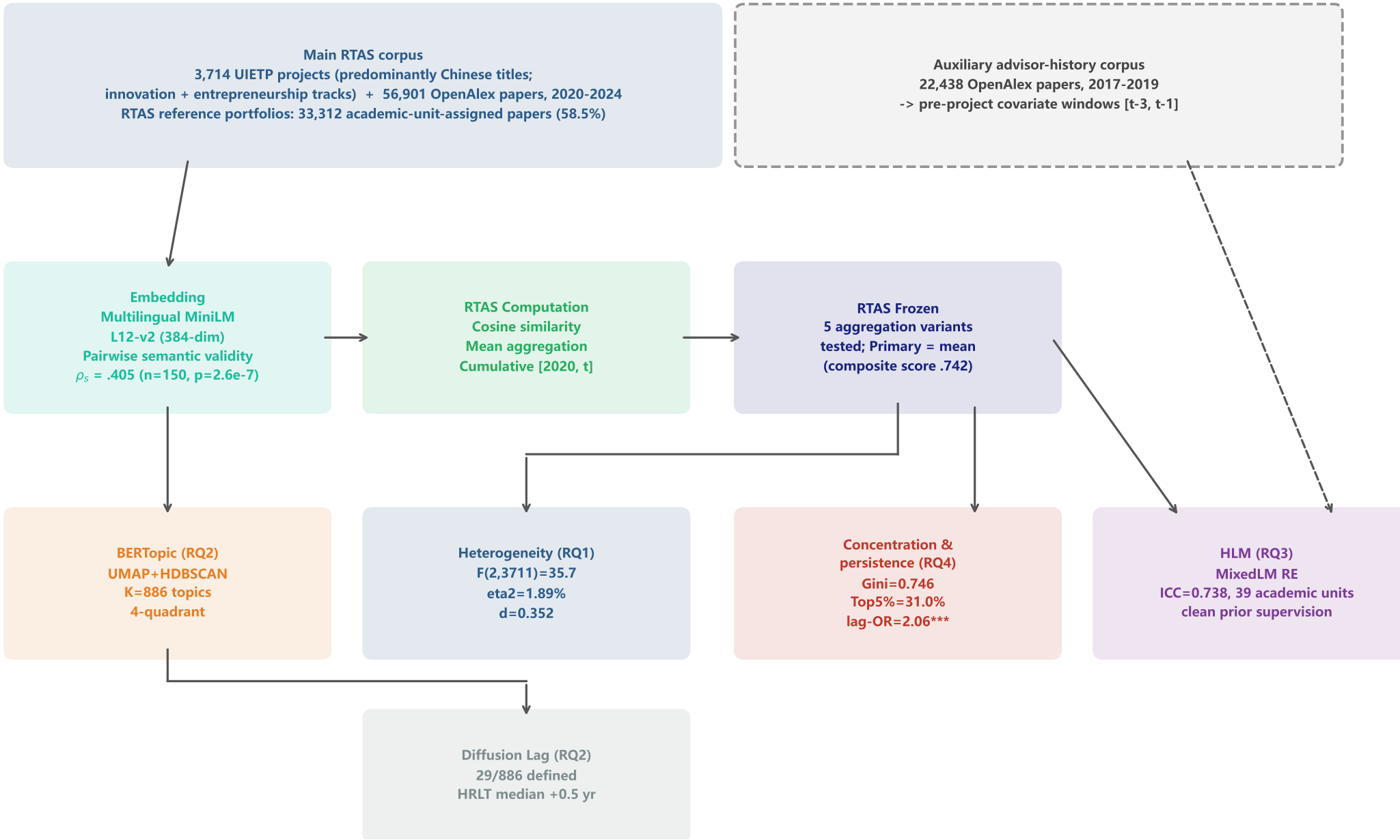
Main RTAS corpus
3,714 UIETP projects (predominantly Chinese titles;
innovation + entrepreneurship tracks) + 56,901 OpenAlex papers, 2020-2024
RTAS reference portfolios: 33,312 academic-unit-assigned papers (58.5%)
Auxiliary advisor-history corpus
22,438 OpenAlex papers, 2017-2019
-> pre-project covariate windows [t-3, t-1]
Embedding
Multilingual MiniLM
L12-v2 (384-dim)
Pairwise semantic validity
$\rho_S$ = .405 (n=150, p=2.6e-7)
RTAS Computation
Cosine similarity
Mean aggregation
Cumulative [2020, t]
RTAS Frozen
5 aggregation variants
tested; Primary = mean
(composite score .742)
BERTopic (RQ2)
UMAP+HDBSCAN
K=886 topics
4-quadrant
Heterogeneity (RQ1)
F(2,3711)=35.7
eta2=1.89%
d=0.352
Concentration &
persistence (RQ4)
Gini=0.746
Top5%=31.0%
lag-OR=2.06***
HLM (RQ3)
MixedLM RE
ICC=0.738, 39 academic units
clean prior supervision
Diffusion Lag (RQ2)
29/886 defined
HRLT median +0.5 yr